ORGANIC GROWTH

The sunrise sector shines bright as India taps into an early-mover advantage

By Amarnath K. Menon

On the eve of Independence Day this year, the L.V. Prasad Eye Institute in Hyderabad had some good news to share. They had, in collaboration with the Centre for Cellular and Molecular Biology and IIT Hyderabad, developed the first made-in-India 3D-printed human cornea that could be an affordable solution for corneal blindness.

This synergy between public and private entities shows how far India has come in the field of biotechnology since the first biotech firm, Biocon, was launched in 1978. Eight years later, a dedicated Department of Biotechnology (DBT) was set up at a time when very few people were working in biosciences.

DBT used the early years to develop skilled human resources out of the many science and tech grads, create infrastructure, promote R&D, and lay the ground for a regulatory framework for the sunrise sector. DBT, therefore, is now at the forefront of enabling cutting-edge research and innovation, with a strong emphasis on translational research. It came up with a National Biotechnology Development Strategy (NBDS) in 2007, to "build coherence and connectivity between disciplines". NBDS-II (2015-20) aims to make India a world-class biomanufacturing hub through the four missions of healthcare, food and nutrition, clean energy and education—and become a $100 billion sector by 2025.

Though pharma, industry, agriculture, infotech and services remain focus areas for biotech, India's emergence as a prominent vaccine-maker globally has given it a lead in the supply of DPT, BCG, measles and Covid vaccines. We also lead in biosimilars, an industry estimated to reach $12 billion in value by 2025. India is also strong in bioservices, and is now home to most FDA-approved plants outside the US. ■

GETTY IMAGES



LOOKING INTO THE FUTURE
Students at the Tata Institute of Fundamental Research, Mumbai, in the '40s

5,000+
NO. OF BIOTECH FIRMS, INCLUDING 4,000-ODD START-UPS, IN INDIA

38,595
NO. OF PUBLICATIONS FROM INDIA, 7.7 PER CENT OF THE TOTAL GLOBAL BIOTECH SCHOLARLY OUTPUT